



驱动创新：英飞凌氮化镓解决方案在机器人中的应用

Driving innovation: Infineon GaN solutions in robotics

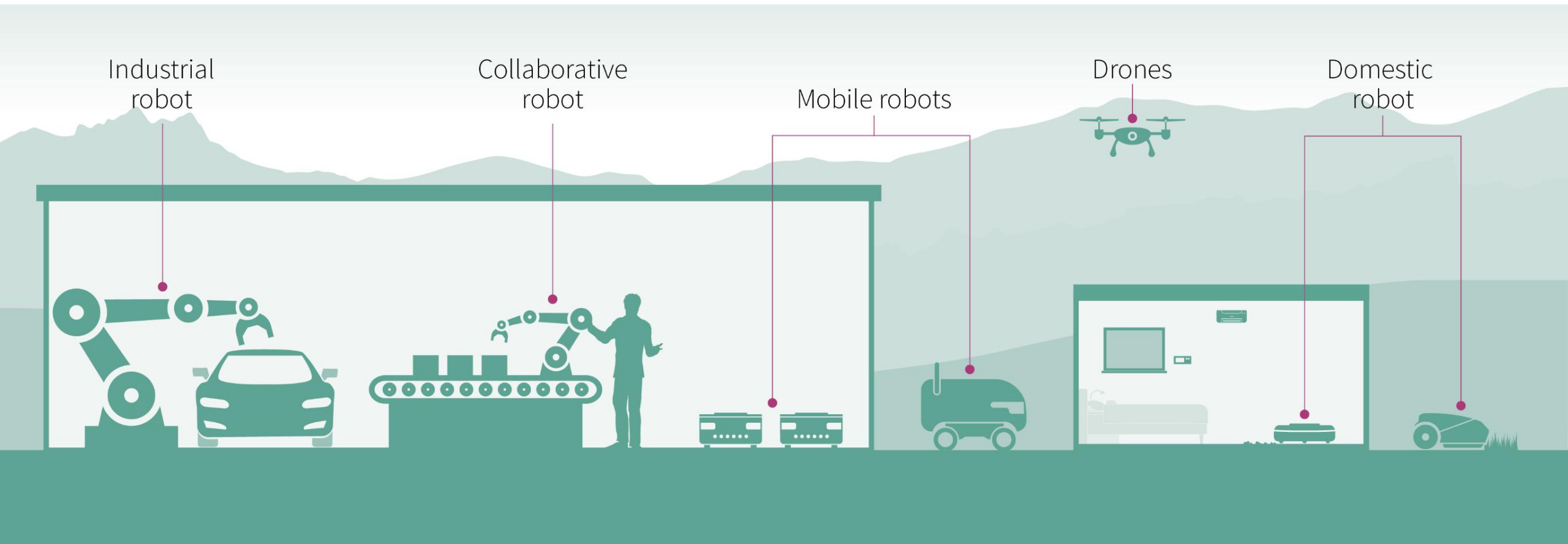
Alan Lun 伦伟强

March 2025 (Shenzhen & Beijing)



Robots are all around us 机器人就在我们身边

机器人在工业环境中为我们提供支持，运送和运输货物以及清洁我们的家园



Robotics are an evolving application

机器人是一种不断发展的应用



已建立机器人类别

协作机器人（工业）
无人机（农业、巡检）
割草机器人
真空吸尘机器人
物流机器人
无人机（消费级）

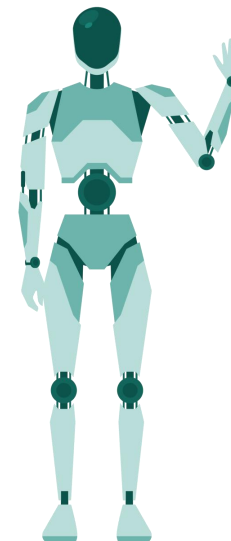
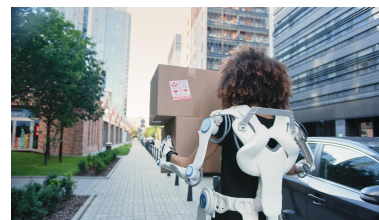
更高水平的自动化

无人机（运输）
酒店机器人
协作机器人（消费者）



协助和复杂的人工任务

人形机器人
外骨骼
物流机器人（室外）



Power density

中压/低压CoolGaN™

低压 FOC 电机驱动器的价值

CoolGaN™的优势……



- ✓ 低导通电阻
- ✓ 基本沒有 Q_{RR}
- ✓ Q_{OSS} 更低
- ✓ C_{OSS} 更加线性
- ✓ Q_G 更低
- ✓ 低寄生电感封装

…启动新的系统值…



- ✓ 更低的开关损耗
- ✓ 更清晰的开关波形
- ✓ 更短的死区时间
- ✓ 更高的开关频率

…带来颠覆性的解决方案!



- ✓ 功率密度提高
 - ✓ 功率段体积更小
 - ✓ 电容体积更小
 - ✓ 更小的散热器
- ✓ 更高效率
 - ✓ 逆变器效率更高
 - ✓ 整体系统效率提升
- ✓ 控制环路带宽更高

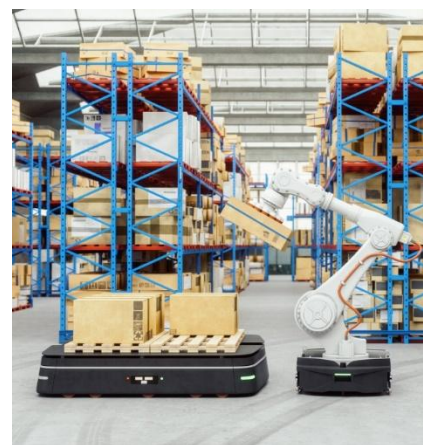
英飞凌CoolGaN™ 在机器人、AGV 和工业自动化中的应用: 推动功率密度和性能的新基准



- CoolGaN™产品组合和配套的电机驱动设计可为您的系统提供最高的W/in³和任何规模的效率，从 20 到 150V_{DC}，从数百瓦到数十千瓦

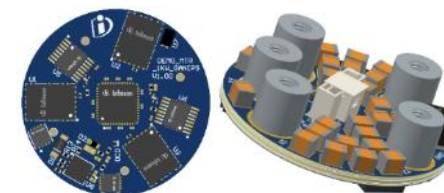
GaN FOC的设计为什么要选择更高电机控制中载波频率(f_{sw})?

- 无需使用笨重的电解电容器 → 利用分布摆放陶瓷电容器组改善尺寸、可靠性和 EMI
- 易于将逆变器集成到电机底盘 → 消除逆变器散热器，减少每个关节/轴的布线，并简化 EMC 设计
- 更高的控制环路频率可提高动态响应

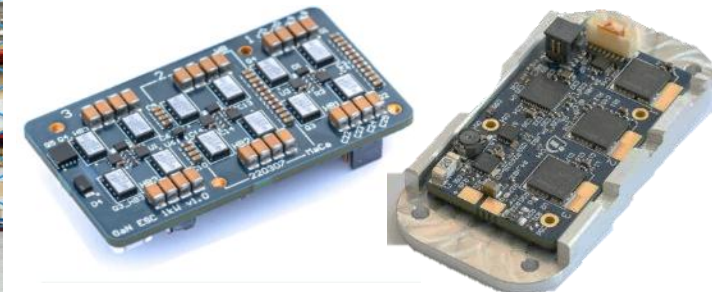


300 W – 1 kW

in 3.4 cm diameter



1-2 kW in 3 x 5 cm

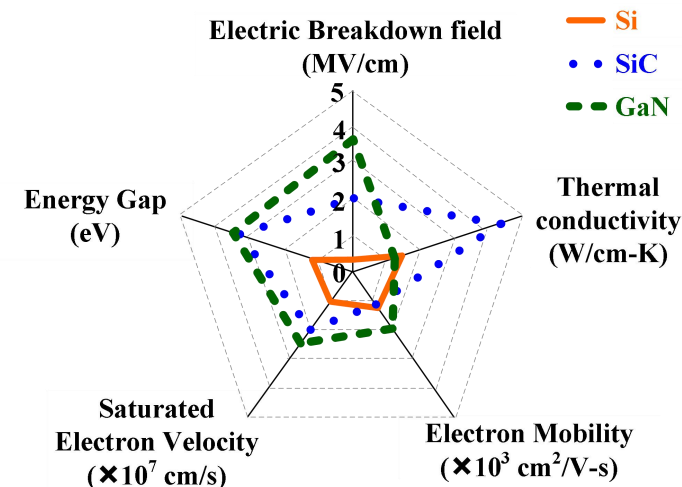


10-20 kW in 5 x 13 cm

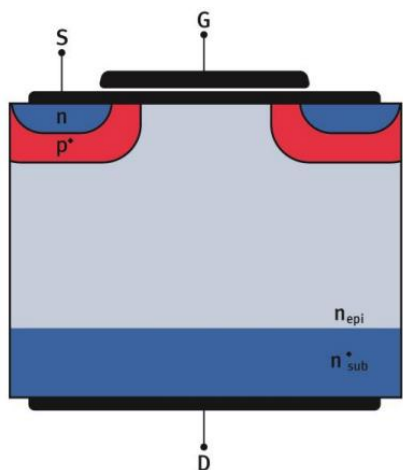


什么是中压CoolGaN™?

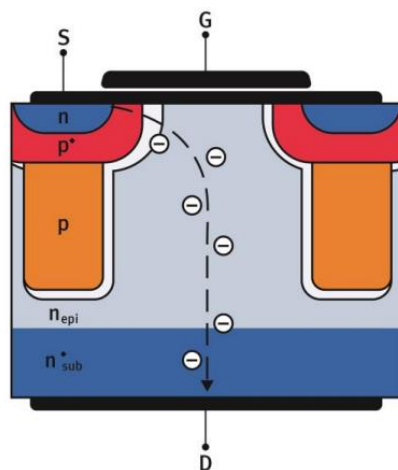
- › 宽禁带(WBG) 半导体 (SiC 和 GaN) 的优异材料特性使其非常适合用于功率晶体管 → GaN 高电子迁移率晶体管(HEMTs)
- › 常关型HEMTs:
 - Cascode与低压Si MOSFET共同封装
 - 增强型栅极注入晶体管(ohmic gate)
 - 增强型肖特基栅HEMT (pseudo-insulated gate)
- › CoolGaN™ HEMTs比Si MOSFET更小、开关速度更快, 且具有不同的封装类型、开关行为和栅极驱动要求
- › 600-700 V 和 40-200 V GaN 之间的差异: 栅极驱动方案可能不同, 低压 GaN 具有较高的 di/dt 和较低的 dv/dt , 从而导致不同的设计考虑



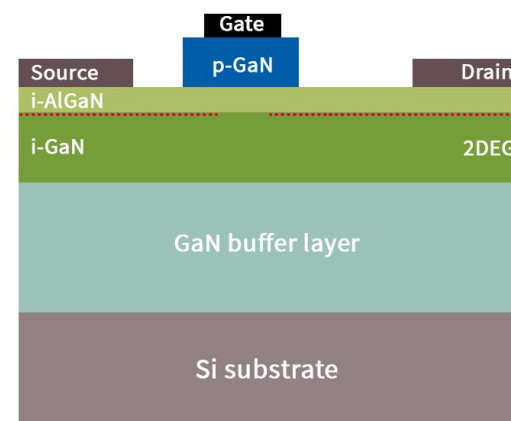
Planar D-MOSFET



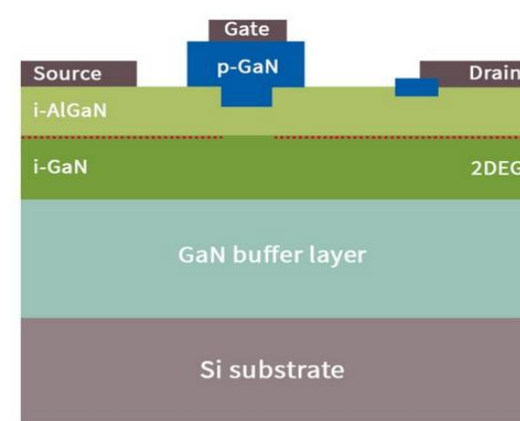
SuperJunction MOSFET



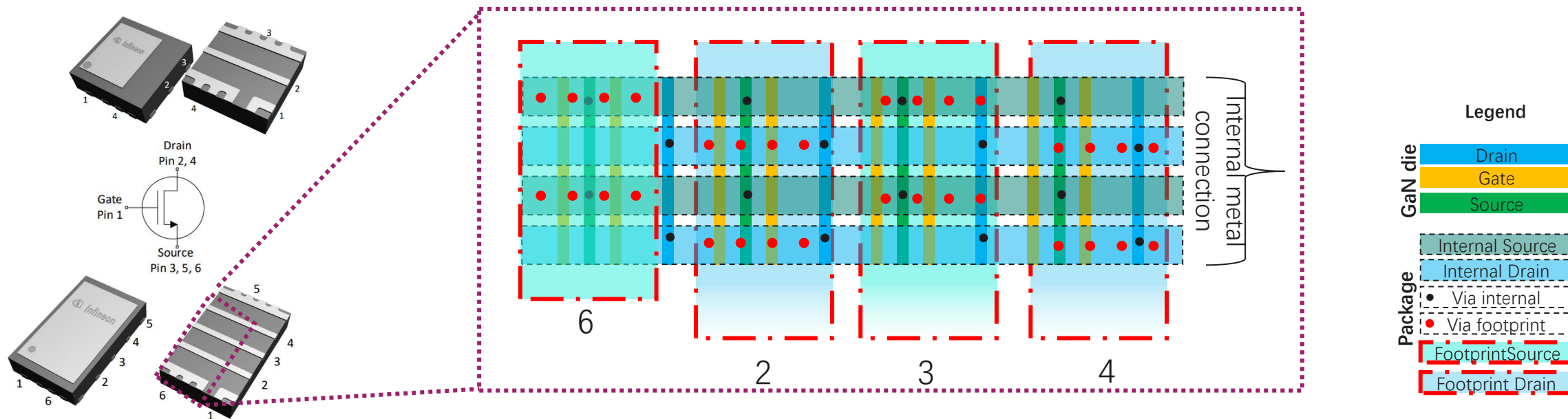
E-Mode SG GaN HEMT



E-Mode GaN GIT

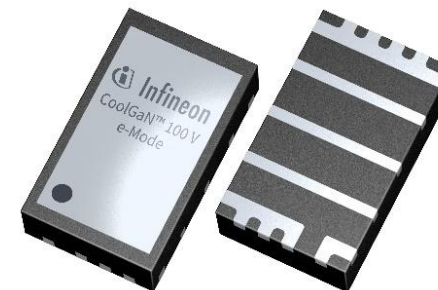
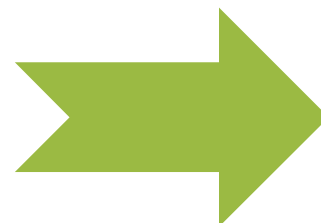


中压GaN TSON-4 (3x3) and TSON-6 (3x5)封装描述



- 英飞凌CoolGaN™ TSON封装采用经济高效且高性能的引线框架技术
- 晶体管由多个并排放置的漏极/源极/栅极指状物组成
- 由于 GaN 的横向特性，所有三个端子（栅极、漏极和源极）都位于芯片的同一侧
- 这些芯片端子通过交错拓扑引到外部
- 此实施的好处：
 - 最低的寄生影响（电阻和电感）
 - 最简单且经济有效的方法
 - 面朝下提供最佳的热连接，直接冷却通道

中压 GaN 封装比较

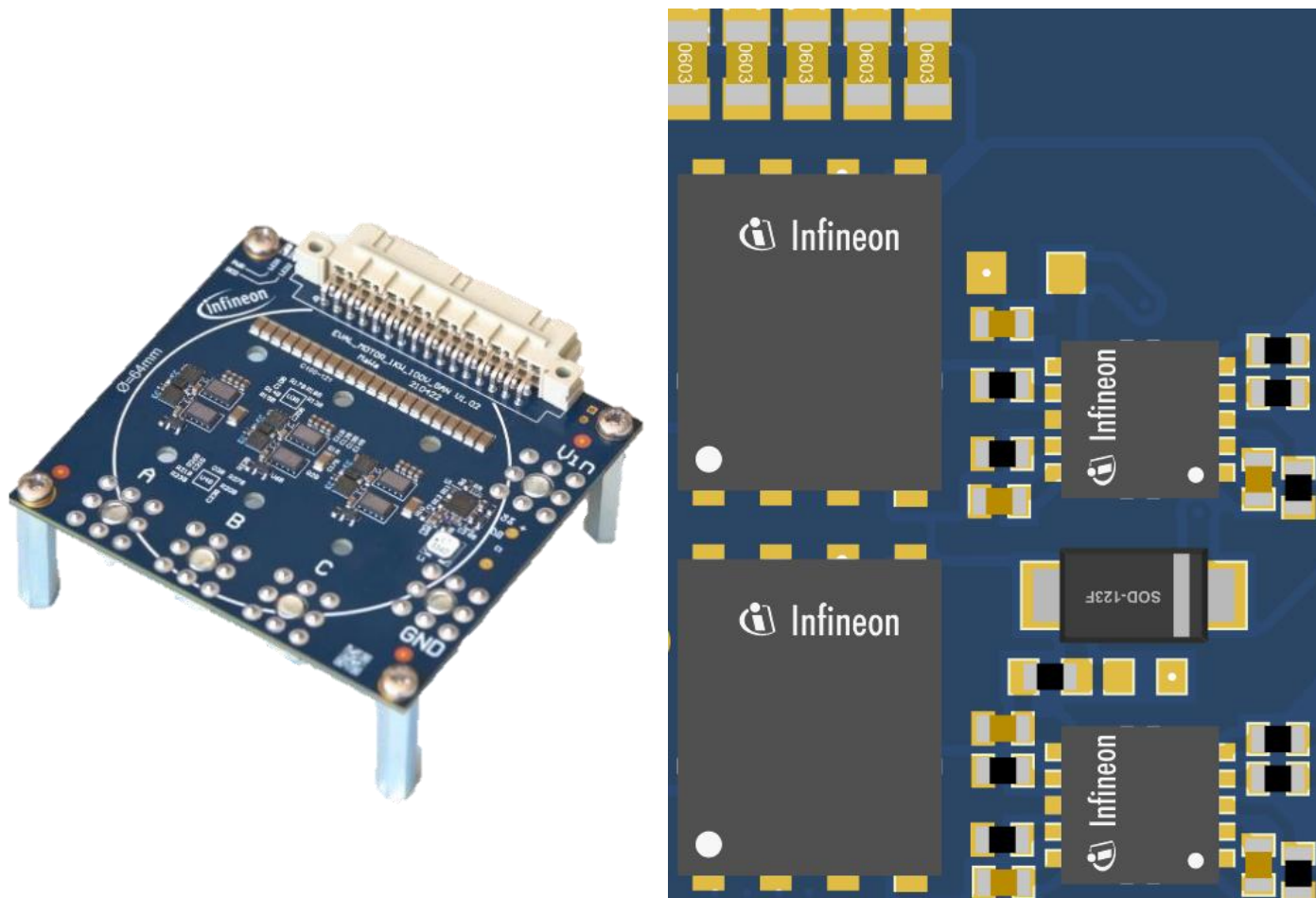


- Half the package size for smaller R_{dson}
 - IGC033S10S1 PQFN **3x5 mm**
 - ISC030NM10NM6 SSO8 **5X6 mm**
- Supreme the package thermal resistances on PQFN 3x5
 - **~30% lower thermal resistance than SSO8 5X6**
- Less than 1/5 gate charge
 - IGC033S10S1 **11nC**
 - ISC030NM10NM6 **55nC**

	100 V OptiMOS™ 6	100 V CoolGaN™
Product Name	ISC030NM10NM6	IGC033S10S1 (ES)
Package	SuperSO8, BSC	PQFN 3x5, DSC
$R_{\text{DS(ON),typ}}$	2.6 mΩ	2.4mΩ
$Q_{\text{oss,typ}}$	101 nC	49 nC
$Q_{\text{rr,typ}}$	56 nC	0 nC
$Q_{\text{g,typ}}$	55 nC	11 nC
V_{DRV}	10 V	5 V
FOM_{G} [mΩ x nC]	143	26
$R_{\text{thJC(TOP),max}}$	20 K/W	0.6 K/W
$R_{\text{thJC(BOTTOM),max}}$	0.72 K/W	2.8 K/W

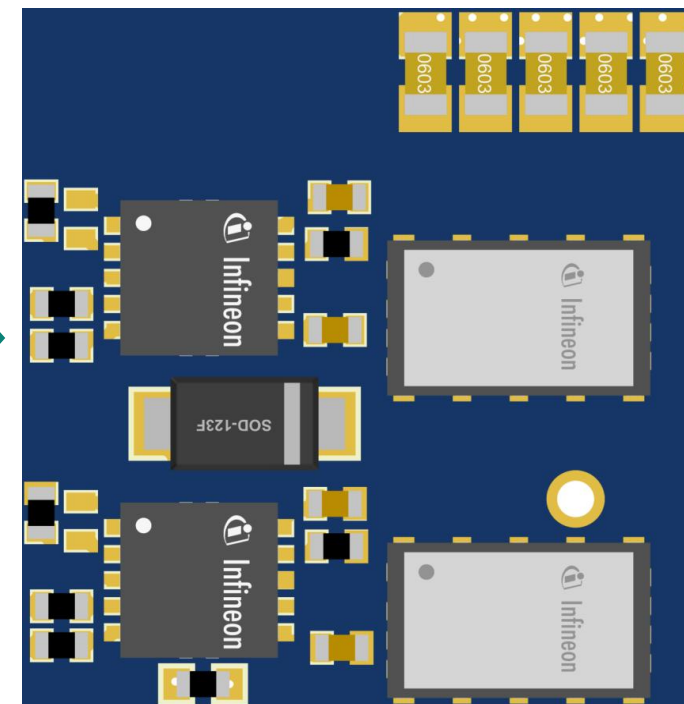
CoolGaN™与OptiMOS™ 6技术比较

针对高速开关转换进行优化的电机驱动设计



OptiMOS™ 6 in 5x6 mm SSO8 with
1EDN7116G driver (220 mm² area)

40% area
reduction



CoolGaN™ in 3x5 mm PQFN with
1EDN7126G driver (140 mm² area)

通过降低电机铁芯损耗来提高最终系统效率

测试条件: 48 V, 308 W, 4000 RPM

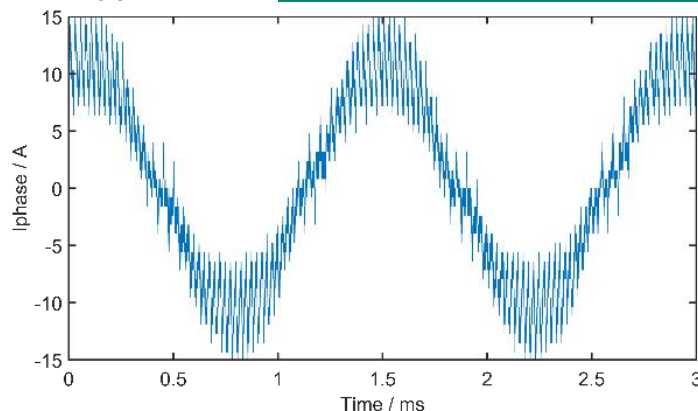
载波频率: 20 kHz

- › 整体系统效率(包括电机): **89%**
- › 电机温度: **68°C**
- › 电机电流纹波: **10 A_{pp}**

Motor runs 13°C cooler

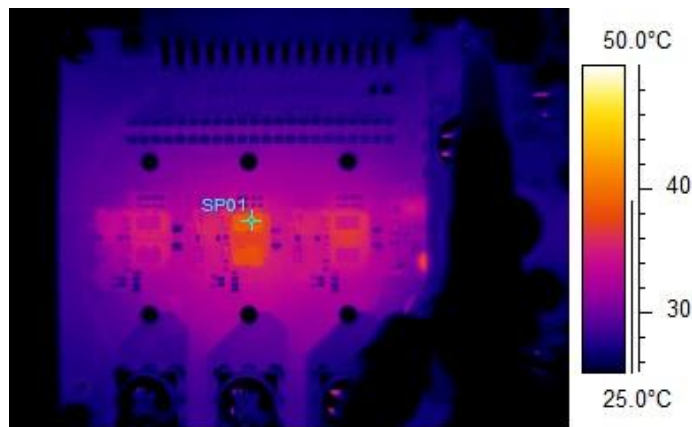
with 5x less torque ripple

电机相电流波形:



CoolGaN™

电路板热像图:

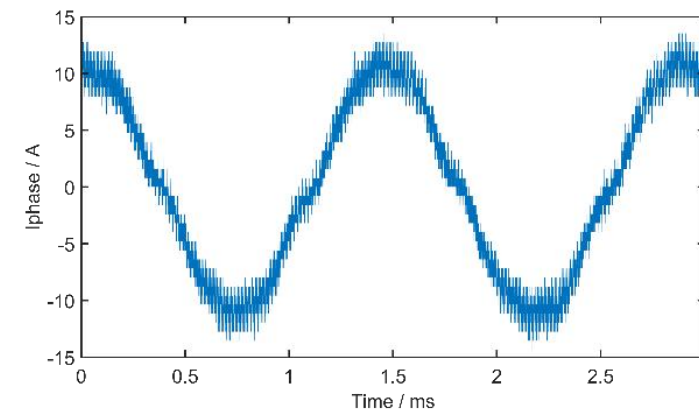


载波频率: 100 kHz

- › 整体系统效率(包括电机): **96%**
- › 电机温度: **55°C**
- › 电机电流纹波: **2 A_{pp}**

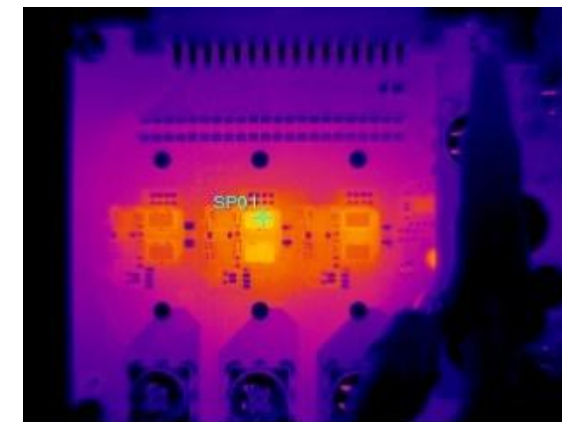
Motor loss	↓↓↓
Inverter loss	↑
System loss	↓↓

电机相电流波形:

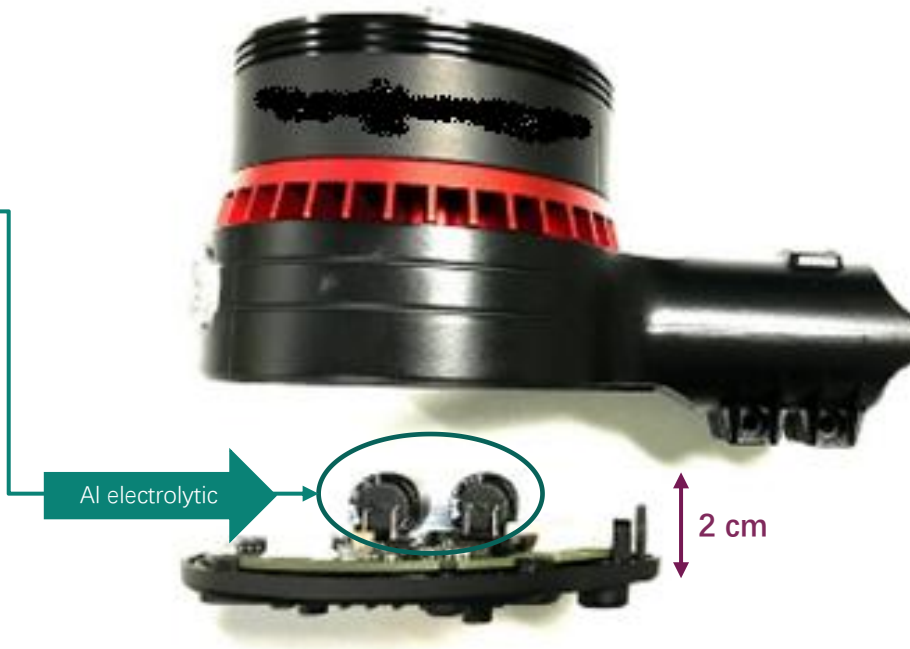
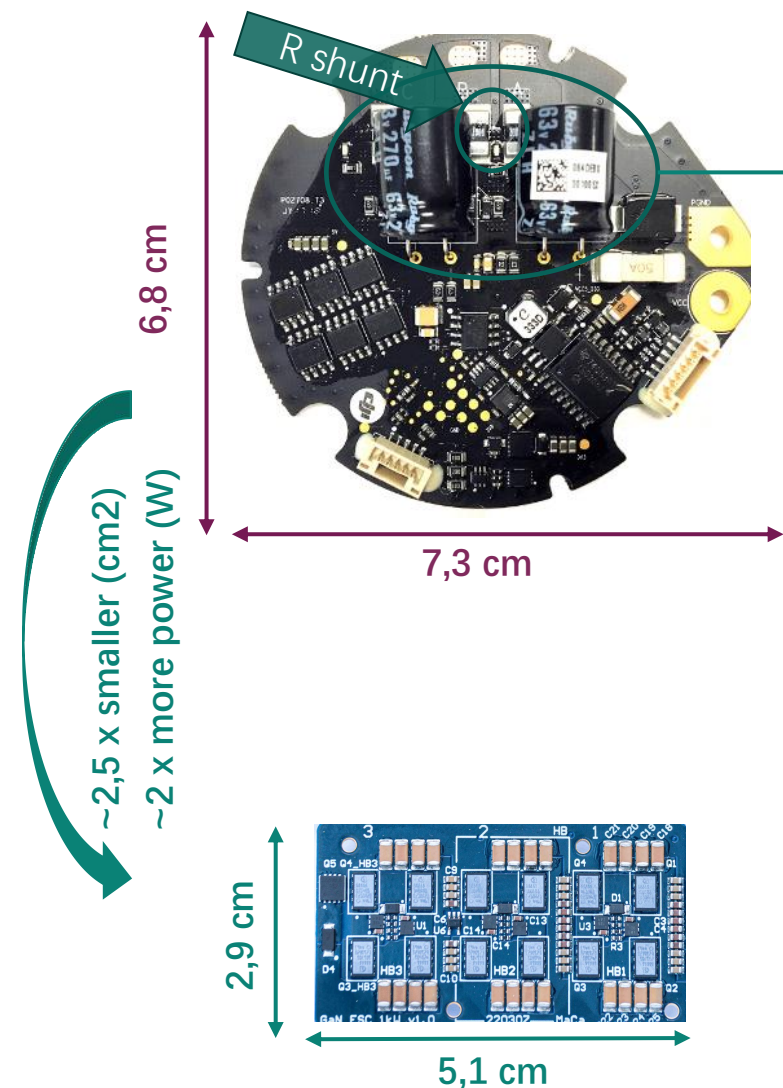


CoolGaN™

电路板热像图:



使用 GaN 将尺寸缩小 2.5 倍



Driver characteristics:

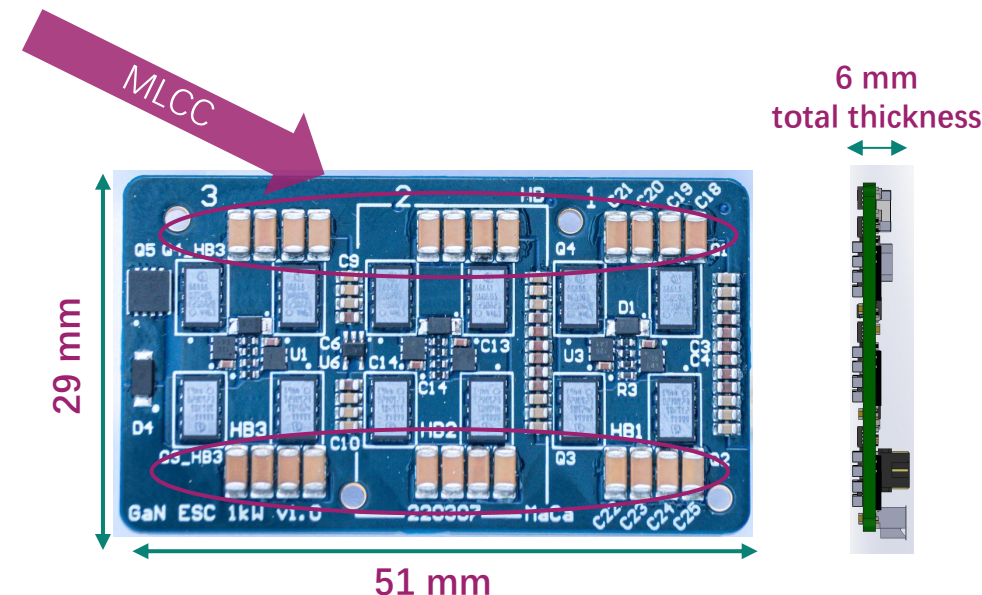
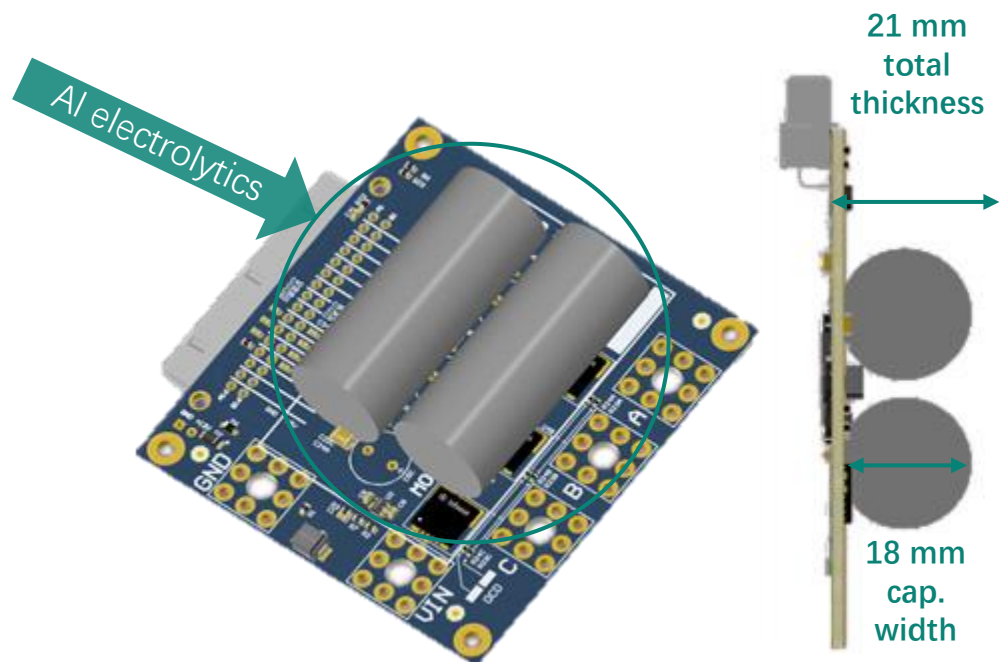
- Max Motor power: **800 W**
- Max Battery current (for one ESC): 17 A
- Battery voltage: 48V
- **fsw=24kHz**
- **Area = 38,5 cm²**
- PCB's layers = 6
- **DC-link= Al electrolytic**
- **Current sens. = R shunt**
- **Bulk capacitance defined by RMS capability**

Driver characteristics:

- Max Motor power: **1,5 kW**
- Max Battery current (for one ESC): 20-30A
- Battery voltage: 48V
- **fsw=100kHz**
- **Area = 14,8cm²**
- PCB's layers = 6
- **DC-link= MLCC**
- **Current sens. = coreless Hall sensor based**
- **Bulk capacitance defined by voltage ripple**



通过提高 FOC载波频率来减小电容器尺寸



OptiMOS™ with $f_{sw} = 20 \text{ kHz}$

1800 μF^* , **11 A RMS current**, <1% voltage ripple, 2000 h lifetime

Bulk capacitance defined by
RMS capability

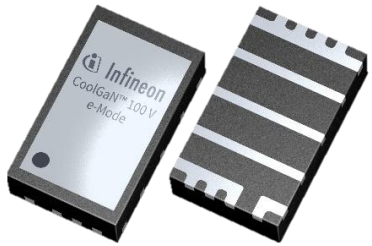
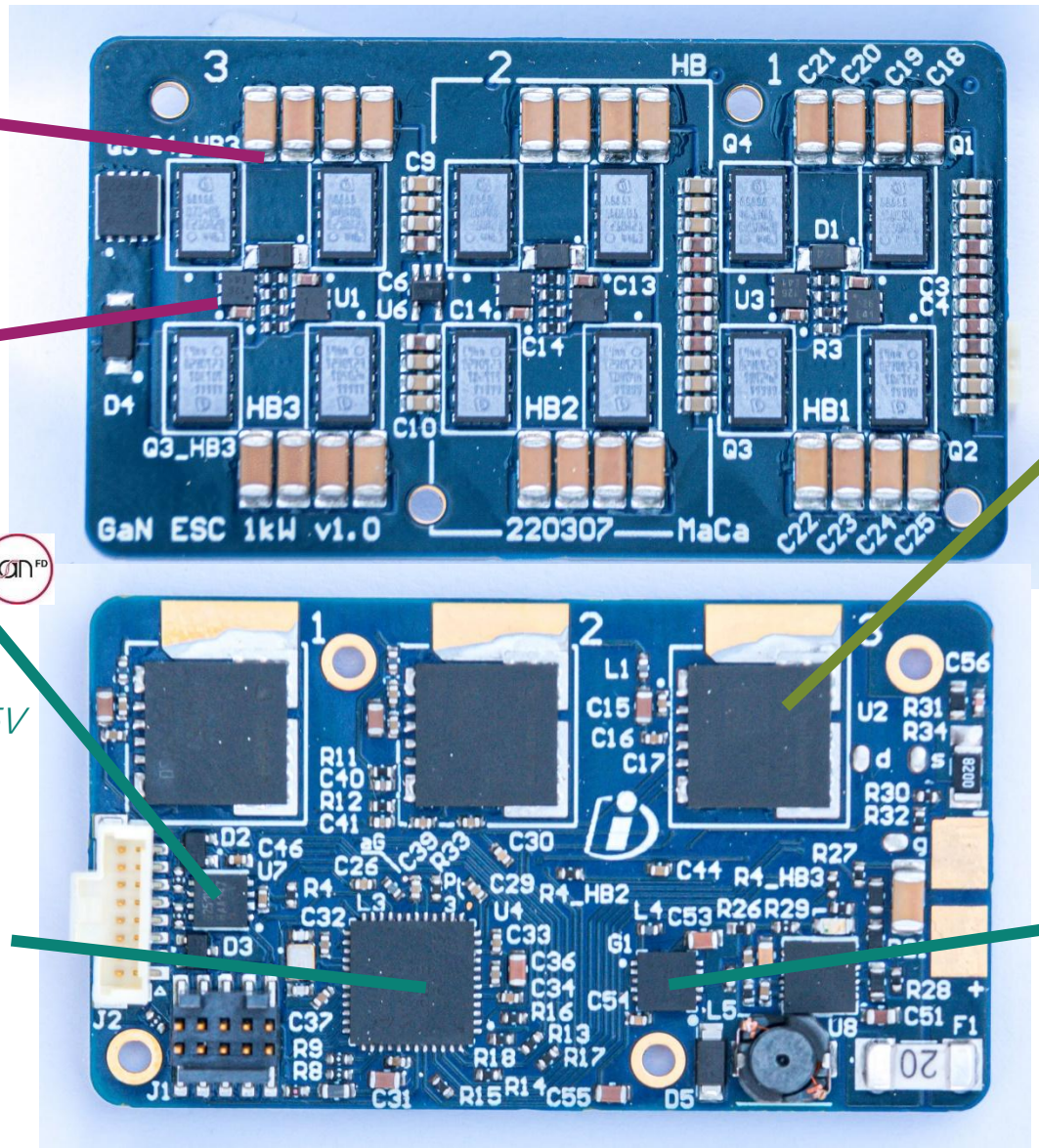
CoolGaN™ with $f_{sw} = 100 \text{ kHz}$

$\geq 88 \mu\text{F}$ (24x 4.7 μF), **>11 A RMS current**, 1% voltage ripple

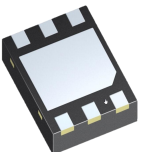
Bulk capacitance defined by
voltage ripple

*Single capacitor is from Utd. Chemi-con, EGPD101ELL911MM40H :Irms~5.5A, Vdc=100V, C=910uF

紧凑的GaN逆变器电机驱动参考设计 - 系统概述



CoolGaN HEMT
100 V, 3 mΩ in 3 x 5 PQFN



1EDN71X6U
High-side 200V with TDI



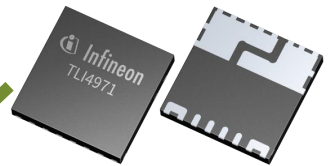
TLE9251
HS CAN Transceiver for 3V3 or 5V



XMC4200
80MHz Cortex M4 + FPU with 150ps HRPWM



PSOC™ Control C3M
(PSC3M5EDLGQ1)
180MHz Cortex M33 + DSP & FPU with 32-bit HRPWM



TLI 4971
Iso. Hall C-sense up to 120A



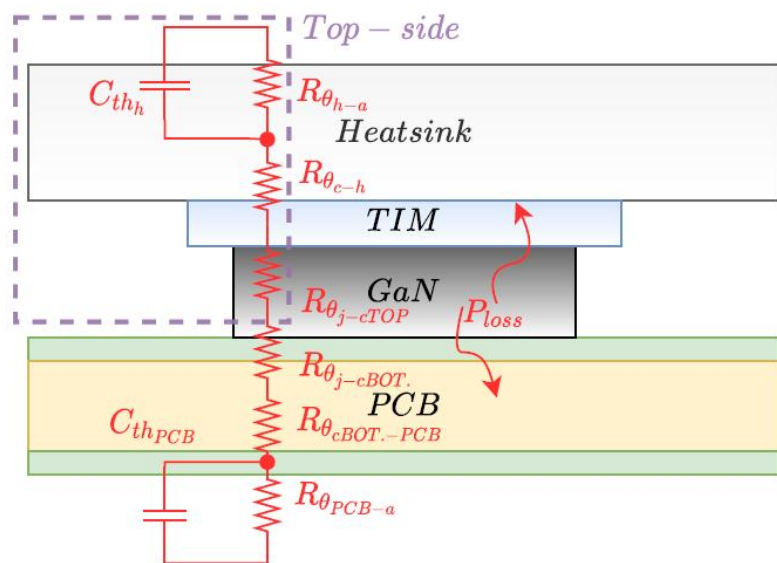
TLI 5570/1/2
TMR Iso. Contactless C-sense up to ±1000A



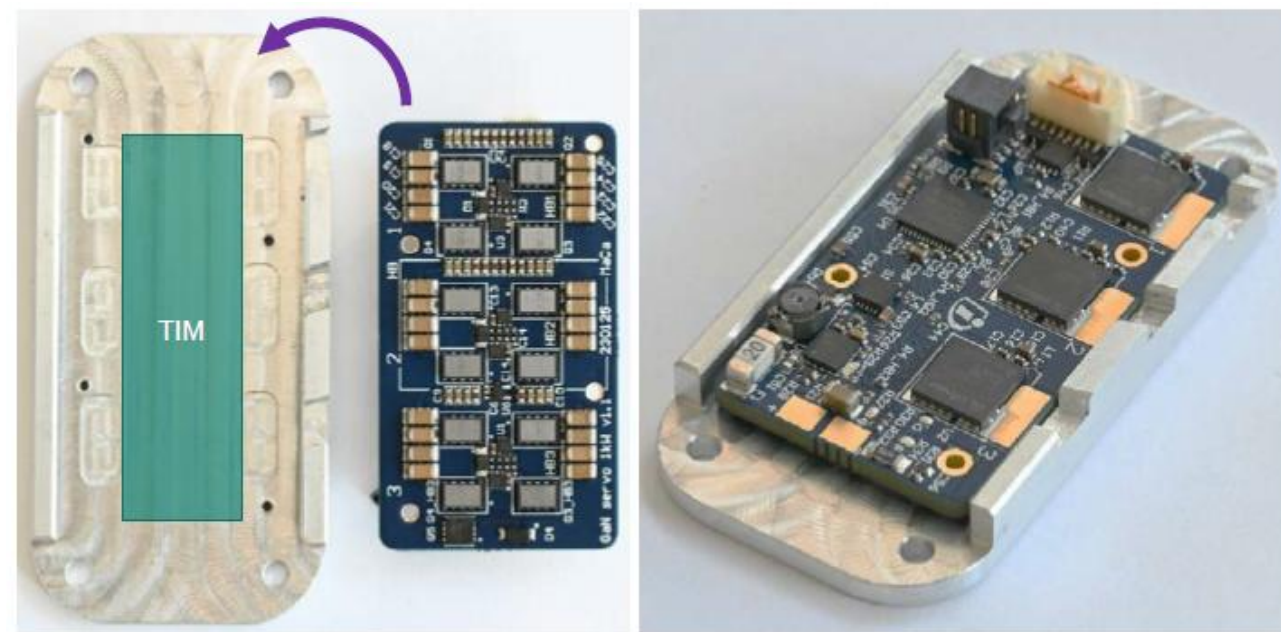
TLS205
Low-noise LDO with 500 mA and OCP / OTP

Size: 29 X 51 X 6 mm
Output current: 34 Arms (with heat spreader and fan)
Output power max: 2kW
Efficiency: 99%

基于TSON-6 (3x5) CoolGaN™ 的热设计

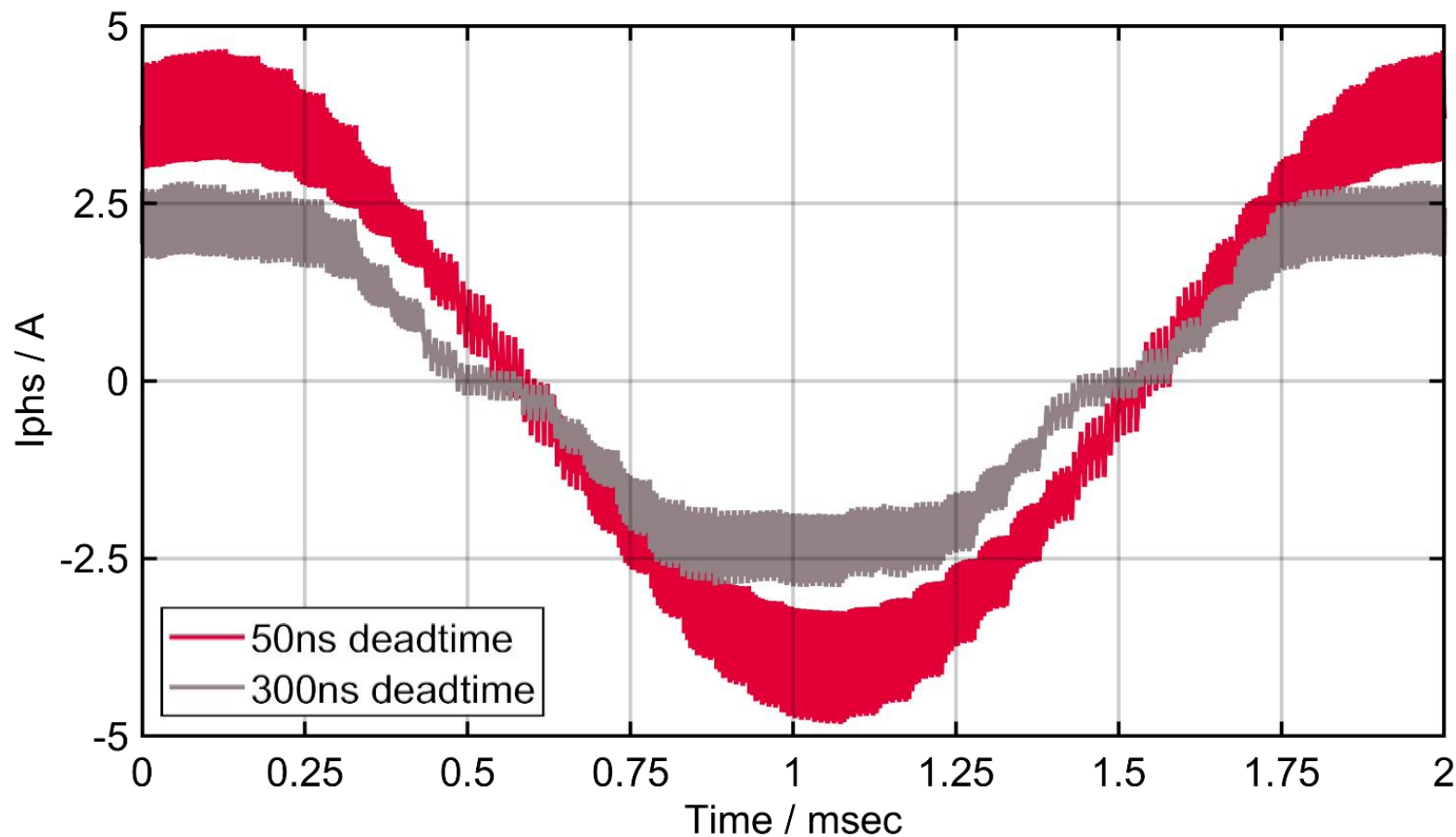


Cross section of the board with a simplified thermal equivalent circuit



Placement of the system board on a heat spreading plate/enclosure

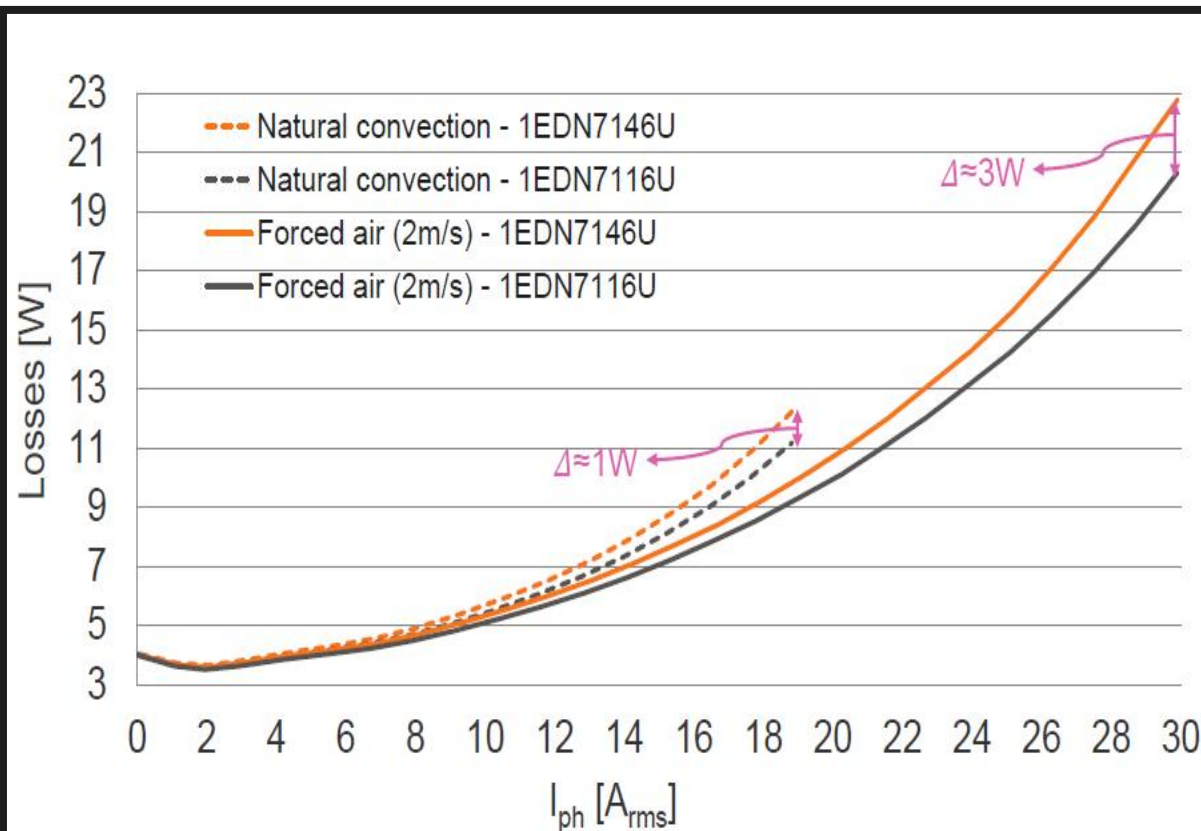
使用 GaN 对死区时间的影响



- CoolGaN™ Transistors可以最大限度地减少由于缺少 Q_{rr} 而导致的死区失真
- 如果死区时间效应在零交叉区域非常明显，并导致失真/振动.
- 此外，较短的死区时间会增加施加到电机的有效电压

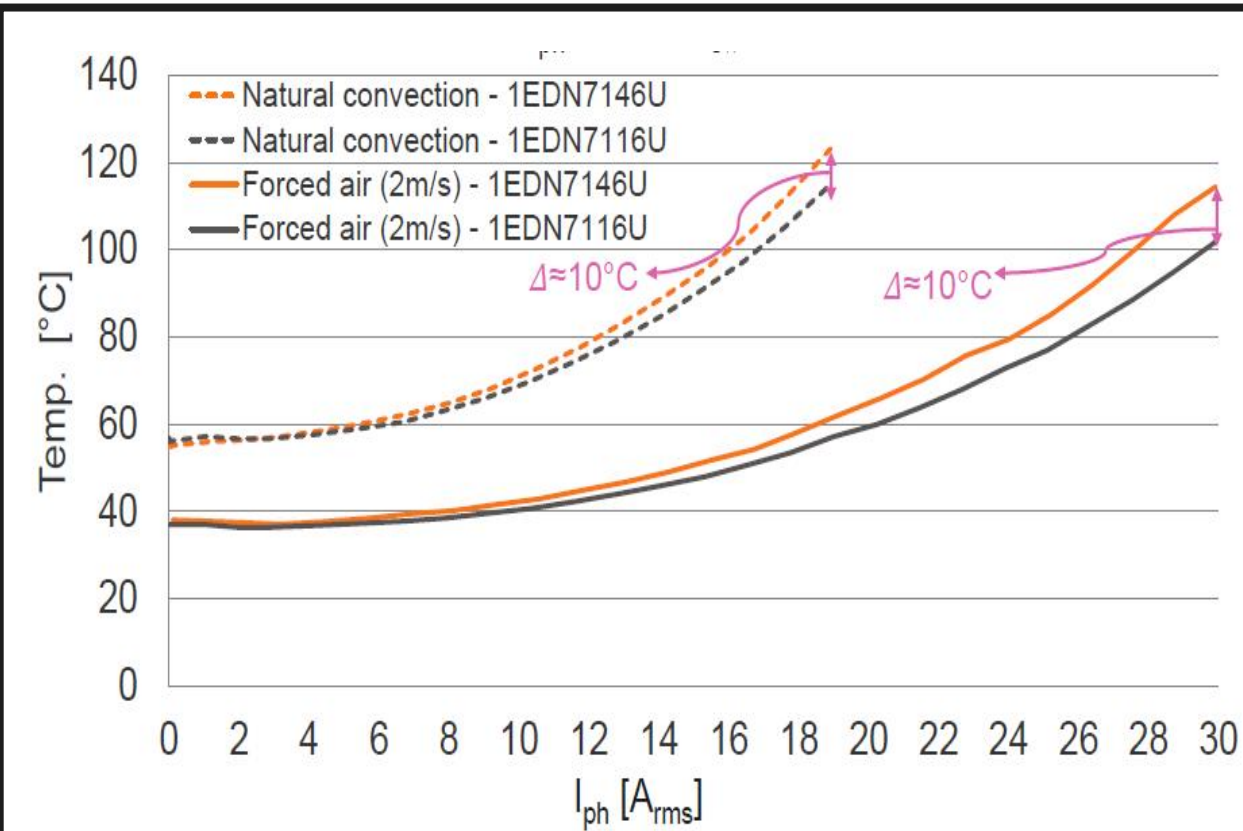
使用 CoolGaNTM 电机驱动板的实验结果

测试条件: $V_{in}=48V$ $t_{dead}=41.2ns$ $L=35\mu H$ $f_{ph}=300Hz$ $f_{sw}=100kHz$



功耗Power dissipation

- 不同栅极驱动器强度和冷却条件下的电机驱动系统损耗



热冷却Thermal performance

- 不同栅极驱动器强度和冷却条件下 GaN 器件的温度

英飞凌中压CoolGaN™在机器人中的应用支持

演示/参考设计

DC-DC

Drives

Humanoid Motor Robotics

Coming soon

MV CoolGaN™ Transistors产品

Family	Sales Code	Voltage class	R _{DS(on)} , typ	Package type (mm)	Package figure	MP time	Positioning
G3	IGC033S101	100 V	2.4 mΩ	PQFN 3x5 BSC		Production	Lower R _{DS(on)} → higher power rating PQFN package → smaller and lower inductance Bottom-side and dual-side cooling options
	IGC033S10S1			PQFN 3x5 DSC		Production	
	IGB110S101		8 mΩ	PQFN 3x3 BSC		Q3'24 ES available	
	IGB110S10S1			PQFN 3x3 DSC		Q3'24 ES available	
	IGB070S10S1		5 mΩ	PQFN 3x3 DSC		Q4'24 ES available	
	IGC019S06S1	60 V	1.4 mΩ	PQFN 3x5 DSC		Q1'25 ES available	
	IGC025S08S1	80 V	1.9 mΩ	PQFN 3x5 DSC		Q1'25 ES available	Lower cost and lower R _{DS(on)} in smaller package size vs. back-to-back Si MOSFETs
	IGC037S12S1	120 V	2.8 mΩ	PQFN 3x5 DSC		Q1'25 ES available	
	IGC090S20S1	200 V	6.7 mΩ	PQFN 3x5 DSC		Q1'25 ES available	
	IGK080B041S	40 V BDS	6 mΩ	WLCSP 1.7x1.7		Q4'24 ES available	
	IGK120B041S	40 V BDS	9 mΩ	WLCSP 1.2x1.7		Q1'25 ES in Q3'24	Higher power density with integrated power stage, easier design-in
	IGK048B041S	40 V BDS	4 mΩ	WLCSP 2.1x2.1		Q1'25 ES in Q3'24	
G1	IGD044H10S1	100 V Drive	4.4 mΩ (x2)	RQFN 4.5x5.5		Q3'25 EES in Q1'25	Chip embedding → Thermally enhanced package Bottom-side or top-side cooling options
	GS61008P	100 V	7 mΩ	CE 7.6x4.6x0.51 BSC		Production	
	GS61008T			CE 7.0x4.0x0.54 DSC		Production	
	GS61004B		16 mΩ	CE 4.6x4.4x0.51 BSC		Production	

针对优化驱动 MV CoolGaN™

1EDN71x6G
3×3 mm

1EDN71x6U
1.8×1.8 mm

2EDL501xxx-U2D (coming soon)
2x2 mm TSNP-12 package

